CSA1718 : Artificial Intelligence

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2.8 Queen's Problem:

```
N = 8
board = [-1] * N


def solve(row):
    if row == N:
        return True

    for col in range(N):
        safe = True

        for r in range(row):
            if (board[r] == col or
                abs(board[r] - col) == abs(r - row)):
                safe = False
```

```
                break

        if safe:
            board[row] = col
            if solve(row + 1):
                return True
            board[row] = -1

    return False

if solve(0):
    for r in range(N):
        print(" ".join(
            "Q" if board[r] == c else "."
            for c in range(N)
        ))
```

Output:

